
Correct Scores, Wrong Scientific Conclusions: Auditing Learned Surrogates of Chaotic Dynamics

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Learned forecasters increasingly stand in for scientific models that are themselves
2 imperfect, and a forecast score alone cannot say whether the surrogate succeeds
3 for the right reasons. We test five conditions that connect a score to a scientific
4 claim (held-out generalization, complete failure accounting, matched disclosure,
5 structural identifiability, and cross-system transfer) on k -pendulum ($k \in \{1, 2, 3\}$)
6 and Lorenz-63 forecasts from language models, a time-series foundation model,
7 learned dynamics, sparse identification, and classical integrators. Every test alters or
8 overturns a conclusion the raw comparison invites: the in-sample/out-of-sample gap
9 is fourfold, two exact symmetries make the apparent parameter recovery impossible,
10 and a second system reverses the model ordering. On video-tracked hardware the
11 roles invert: the RK4 model with published constants is itself imperfect, and its
12 lead over simple baselines is gone within a second. A correct leaderboard can still
13 support incorrect scientific conclusions.

14 1 Introduction

15 Scientific claims require more than low forecast error. When a learned forecaster stands in for an
16 imperfect scientific model, a result may depend on training trajectories, dropped failures, unmatched
17 inputs, an unidentifiable target, or one favorable system: failures of the evaluation rather than of the
18 surrogate. We study these failure modes directly. The task is to predict a chaotic system’s state s_T
19 from its state s_0 , on two systems chosen to disagree: the k -link pendulum, chaotic for $k \geq 2$, with
20 high-precision ground truth and control over what physical metadata a model is told, and Lorenz-63,
21 which differs in observation model and hypothesis class; a hardware replication of the audit (§4.5)
22 then scores the reference model itself as the imperfect one.

23 Our contribution is a claim audit for dynamical-system forecasts: held-out generalization, complete
24 failure accounting, matched disclosure on identical trajectories, identifiability checked before scoring
25 recovery, and transfer to a second system. Table 1 states the conclusion before and after each test;
26 all five alter or reject a conclusion the unaudited comparison supports. These are conclusions the
27 field actually draws: zero-shot forecasters are routinely scored without failure accounting or matched
28 controls [Gruver et al., 2023, Zhang and Gilpin, 2025], and our setting mirrors that practice. Three of
29 the checks are the familiar concerns of imperfect models applied to the evaluation itself: misspecifi-
30 cation (a Hamiltonian prior on damped dynamics), identifiability (exact parameter degeneracies), and
31 transfer (a hypothesis class containing one system and not the other).

32 Appendix D situates the audited model families and the identifiability and selective-prediction
33 background.

Table 1: Conclusions before and after the claim audit.

Test	Unaudited conclusion	Audited conclusion
Failures	Answered forecasts represent model performance.	Incomplete coverage lowers reliability-adjusted performance.
Generalization	Learned dynamics forecast accurately.	At this training budget, error rises about fourfold and simple baselines perform better.
Disclosure	Withholding constants improves forecasts.	Matched, the effect is small and model-dependent (+0.069 rad pooled).
Identifiability	Absolute constants can be recovered from angles.	Two exact symmetries make absolute recovery impossible.
Transfer	The pendulum order ranks model families.	Lorenz-63 produces a different order.

2 Benchmark Design

2.1 Task and systems

The pendulum state is $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$; hiding its constants does not change the trajectory, which permits the matched test in §4.2. Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t = 10^{-3}$ s, recorded every 0.01 s plus a 10 s pre- $t=0$ context for the time-series models. Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed; because the labels come from RK4, the coarser-step RK4 baseline is a reference by construction, not a competitor.

2.2 Regimes and grid

Three regimes test generalization: normal ($g=9.81$, undamped), changed_disclosed ($g=3.71$, damped), and changed_hidden ($g=1.62$, damped, constants withheld; full values in Table 9). An evaluation cell is a tuple (model, k , regime, horizon), with horizons $\{0.01, 1, 10, 60\}$ s. The LLM grid also varies modality and prompting; all comparisons here use the coordinate, direct setting.

Out-of-sample protocol. Learned models are evaluated off their training data: a shared held-out set (a separate seed) of 180 initial conditions (60 per regime, $k \in \{1, 2, 3\}$) is scored identically for every family. The cross-family comparison uses horizons $\{1, 10\}$ s, where chaos has begun but error has not saturated ($180 \times 2 = 360$ cells per model); in-sample error appears once, to quantify the generalization gap (§4.1).

2.3 Inputs and grouping

Each family sees different information, and we treat that as part of the experimental record (Table 3, Appendix B). The integrators get the exact equations and constants; the learned models get target-system training plus the constants (normal-regime priors in hidden cells); the LLMs get a text description with the constants except in hidden cells; Chronos gets a 10 s pre- $t=0$ history window and no constants at all. The LLMs are large proprietary models while the learned models are small MLPs (Appendix B); larger physics-informed networks are out of scope. We separate known-physics methods (RK4, Euler, symplectic), which serve as numerical references, from the black-box methods being compared.

2.4 Metrics

The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over links and trajectories. A cell is answered only if the model returns a parseable, finite prediction; because scoring answered cells alone would reward abstention on hard cells, unanswered cells receive the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform guess), so answering is part of the score.

Verification record. Each cell retains its identifiers, prompt, raw response, parser outcome, prediction, token counts, and score. Local methods are reproducible from the released checkpoints and seeds; hosted deployments can change or be withdrawn, so for them the released per-cell responses are the record that keeps the evaluation auditable.

71 **Disclosure gap.** We define $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate a penalty
 72 for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty,
 73 §4.2 also toggles disclosure on identical trajectories.

74 3 Methods

75 **LLMs.** We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through
 76 an OpenAI-compatible API (hosted deployments, June 2026), with a structured prompt and JSON
 77 schema (Appendix A); each cell is one temperature-0 call with uniform budgets, no best-of- N .

78 **Time-series foundation models.** We run Chronos-2 [Ansari et al., 2024] locally in its standard
 79 history-conditioned form: a 10 s pre- $t=0$ context of all $2k$ channels forecasts to the target horizon, in
 80 univariate and multivariate modes (the paired mode comparison uses 20 trajectories per cell).

81 **Learned dynamics.** Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and
 82 integrate at inference with `scipy`, comparing derivative matching, rollout, and mixed- k rollout
 83 objectives (Appendix B). The primary models train on nine trajectories, one per (k, regime) cell;
 84 a nested budget study trains fresh checkpoints at $\{1, 5, 20\}$ trajectories per eligible cell with three
 85 initializations (HNN: undamped cells only, since damping violates Hamiltonian conservation).

86 **Integrators.** Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations
 87 of motion; their error is from discretization alone.

88 **Baselines.** Four reference baselines use the same grid: persistence ($\theta_T=\theta_0$), constant velocity
 89 ($\theta_T=\theta_0+\omega_0 T$), a small-angle linearized model, and a nearest-neighbor analog forecast built from
 90 the learned models’ training data. On both held-out seeds (equal-weighted; §4.3) we also fit SINDy
 91 (degree-two, STLSQ) and NVAR (two delays, degree-two, ridge); divergent rollouts are failures at
 92 $\pi/2$.

93 **Lorenz replication.** A secondary Lorenz–63 test compares numerical references, persistence,
 94 SINDy, NVAR, a Neural ODE trained on 20 seed-42 trajectories, and zero-shot Chronos-2 on five
 95 five-trajectory held-out seeds, reporting training-normalized coordinate MAE at $\{0.1, 0.5, 1.0\}$ s
 96 (configuration in Appendix C). HNN and LNN are omitted because Lorenz–63 is dissipative and
 97 non-Hamiltonian.

98 4 Results

99 4.1 Failure accounting and held-out generalization

100 Every model is scored on the shared held-out grid of 360 cells (180 trajectories, horizons $\{1, 10\}$ s);
 101 Table 2’s Answered column gives the fraction with valid predictions. `grok` answers 82% and `lnn`
 102 94%; their ranks fall when missing cells receive $\pi/2$, and a mean over answered cells alone would
 103 flatter both. Intervals are 95% bootstrap over trajectories, excluding LLM per-call sampling variation.

104 Table 2 ranks the models under the access conditions of §2.3; the ranking is conditional on those
 105 inputs, this training budget, and this system (§4.4 tests transfer). The known-equation integrators sit
 106 at the top as references; every learned dynamics model lands closer to the $\pi/2$ baseline than to the
 107 RK4 and symplectic references. The lowest-error black-box entry, the LLM `kimi` (0.538 rad, 99%
 108 answered), has a marginal interval excluding the nearest baseline’s: a statement about one recorded
 109 June 2026 deployment, not about model families, and the Lorenz test rejects the family reading.

110 After equalizing the hidden-regime inputs, nearest-neighbor lookup (0.725) and the linearized pen-
 111 dulum (0.855) beat every learned model, and persistence (1.004) beats all but HNN-rollout (0.972),
 112 though those two intervals overlap. The stable learned models score about 0.26 rad on their training
 113 trajectories and 0.97–1.14 rad held-out, so in-sample evaluation understates error about fourfold.

114 SINDy (0.883 rad) and NVAR (1.248) appear only on the two-seed replication subset (Table 4);
 115 SINDy lands between Chronos and the learned models, NVAR near constant velocity.

Table 2: Out-of-sample cross-family comparison (360 cells, horizons $\{1, 10\}$ s): reliability-adjusted mean angle error (rad), unanswered cells at $\pi/2$, 95% bootstrap CI. RK4 generates the data (reference). †: configurations first evaluated in this work; ‡: simple baselines.

#	Model	Err	95% CI	Answered	Family
–	rk4 (reference)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi†	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni†	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor‡	0.725	[0.657, 0.797]	100%	baseline
7	linearized‡	0.855	[0.774, 0.939]	100%	baseline
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
10	hnn-rollout	0.972	[0.898, 1.047]	100%	learned
11	persistence‡	1.004	[0.934, 1.078]	100%	baseline
12	neural-ode-rollout-mixed†	1.036	[0.967, 1.104]	100%	learned
13	hnn-rollout-mixed†	1.066	[0.990, 1.143]	100%	learned
14	neural-ode	1.110	[1.039, 1.181]	100%	learned
15	neural-ode-rollout	1.121	[1.054, 1.190]	100%	learned
16	hnn	1.138	[1.059, 1.218]	100%	learned
17	constant-velocity‡	1.319	[1.241, 1.396]	100%	baseline
18	lnn† (full-scale)	1.356	[1.282, 1.430]	94%	learned

4.2 Matched disclosure and identifiability

On the held-out set the hidden regime shows lower error than the disclosed (-0.390 rad, $p_{\text{Holm}} < 10^{-3}$, unpaired) and normal (-0.524) regimes; but the regimes use different trajectories, and the hidden ones are slower and less divergent. Toggling disclosure on identical trajectories for the three LLMs (900 paired cells on a separate matched grid; Table 8) reverses the pooled effect: hiding the constants increases error ($\Delta_{\text{disc}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$), helping at $k=1$, reversing slightly at $k=3$; the unadjusted advantage is a difficulty confound.

The disclosure test also has a structural limit: the angular equations are unchanged by $(m, d) \mapsto (am, ad)$ and $(L, g, d) \mapsto (bL, bg, b^2d)$, so for $k=1$ only g/L and $d/(mL^2)$ are identifiable from an angle trajectory, and absolute g , L , m , d cannot be scored as independently recoverable. A single-state constant-naming probe mostly returned Earth-standard defaults (a prior, not recovery; transcripts in the released records); a released check verifies both invariances to below 10^{-13} .

4.3 Replication across seeds and training budgets

Rerun on an independent held-out seed (five trajectories per cell versus 20; Table 4), the selected ordering is unchanged: numerical references lowest, Chronos-2 before the learned models, HNN-rollout best among them. Two seeds do not estimate seed-level uncertainty. The LNN’s low primary-seed coverage was a hardware artifact: its per-step Hessian exhausted the GPU on long rollouts. A CPU rerun recovers the dropped cells ($66\% \rightarrow 94\%$; both seeds use it); the remaining failures are genuine divergences, scored at $\pi/2$, and the two-seed mean stays 1.366 rad.

Training-budget check. Across three initializations and both held-out seeds, Neural ODE error falls from 1.090 at one trajectory per cell to 0.861 at twenty, and normal-regime HNN error from 1.269 to 0.981 (Table 10). The rows cover different regimes; more data helps the Neural ODE monotonically, the HNN plateaus by 15 trajectories, and neither closes the gap to known physics.

4.4 Cross-system test: Lorenz–63

We test whether the pendulum conclusions transfer to Lorenz–63, a chaotic system with a polynomial vector field. They do not. The identifiability verdict flips: on the pendulum the physical constants are not recoverable from an angle trajectory (§4.2), while on Lorenz, SINDy recovers the governing equations themselves, not merely their forecasts. The refit selects exactly the true sparsity pattern and returns $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ to coefficient error 2.2×10^{-13} (Appendix C); forecast error

145 stays below 2×10^{-5} in every seed. The difference is the setting, not the audit: Lorenz-63 lies inside
 146 SINDy’s hypothesis class and is fully observed; the pendulum’s absolute constants are unidentifiable
 147 from angles under any method. Identifiability is a property of the observation model and the
 148 hypothesis class, and the check makes it explicit before recovery is scored.

149 The model ordering also changes. Trained on one seed and evaluated on five held-out seeds (25
 150 trajectories), a Lorenz Neural ODE reaches normalized MAE 0.014 ± 0.008 , 0.079 ± 0.030 , and
 151 0.277 ± 0.063 at 0.1, 0.5, and 1.0 s (mean $\pm$ SE over seeds), below both Chronos-2 modes at every
 152 horizon (multivariate 0.101 ± 0.018 , 0.691 ± 0.129 , 0.771 ± 0.097 ; univariate similar); NVAR drifts
 153 to 0.429 by 1.0 s and persistence trails (0.900; Table 4). This reverses the pendulum, where trained
 154 models lost to trivial baselines: here the system-trained model beats the zero-shot foundation model.
 155 The families again see different inputs, so this is not an architecture race; what the cross-system result
 156 rejects is the pendulum ranking as a general ordering of model families.

157 The reversal does not rest on a favorable seed: the Neural ODE beats both Chronos-2 modes in all
 158 15 seed-by-horizon comparisons, and the identifiability half is not a sampling claim at all, since the
 159 symmetries are exact and Lorenz-63 lies in SINDy’s library by construction. The test excludes the
 160 API LLMs, HNN, and LNN (a Hamiltonian surrogate is misspecified for the dissipative flow); those
 161 omissions limit what it can establish, not what it can refute, and the Lorenz ordering is not offered as
 162 the correct one.

163 4.5 Real data: the reference model becomes the imperfect one

164 On real hardware the RK4 integrator is no longer ground truth but the imperfect model. We use an
 165 open video-tracked double pendulum [Myers et al., 2020] (three free-drop trials, 500 Hz, published
 166 point-mass constants, $\sim 0.35^\circ$ median tracking noise); Trial 1 only calibrates (nearest-neighbor
 167 database, SINDy fit, one-parameter damping fit), and trials 2–3 supply 120 held-out states scored as
 168 before at $\{0.1, 0.5, 1.0\}$ s (Table 5, Appendix E).

169 RK4 with the published constants leads at 0.1 s (0.084 rad, about $14\times$ the tracking noise) but reaches
 170 1.25 rad by 1.0 s, indistinguishable from persistence (1.23); unmodeled friction and the point-mass
 171 simplification consume its advantage within a second. The fitted damping trades horizons rather than
 172 dominating (0.89 vs 0.82 at 0.5 s, 1.15 vs 1.25 at 1.0 s). SINDy, exact on Lorenz, sits at baseline
 173 level here (1.54); the hardware’s dynamics are not in its library. Chronos-2 answers 89% of cells,
 174 since states shortly after the drop lack the 5 s context used here (Appendix E); failure accounting
 175 charges them at $\pi/2$. Every check transfers, and the ordering differs from both synthetic systems.

176 4.6 Diagnostics

177 Appendix F reports the remaining diagnostics. Two matter most: the physical-fidelity check localizes
 178 a misspecification (a symplectic HNN rollout leaves energy drift nearly unchanged, so the fault is the
 179 learned Hamiltonian, not the integrator), and every black-box family saturates near the chaotic noise
 180 floor by 1 s, hence the $\{1, 10\}$ s window. Error rises with k for all but one model; the hidden regime
 181 is easier for most models (the §4.2 confound); rollout loss is not uniformly better; and Chronos-2’s
 182 two modes are indistinguishable on 720 paired samples.

183 5 Discussion and Conclusion

184 Performance depends on system, observations, hypothesis class, data, and access; a scalar leaderboard
 185 does not encode those conditions, and each of the five checks changes a claim the raw leaderboard
 186 supports. Benchmarks should report the checks, training budgets, and information access as part
 187 of the result. The synthetic systems make each check’s effect visible against exact ground truth;
 188 the hardware check runs the same audit where the reference model itself is imperfect. All prompts,
 189 outputs, checkpoints, seeds, and compute records accompany the paper.

190 **Limitations.** The LLM grid uses one held-out seed, the replication a smaller second, the budget
 191 study three initializations, the Lorenz test five trajectories per seed (no API LLMs, HNN, or LNN),
 192 and the hardware check three trials of one apparatus with filtered velocities; SINDy needs the
 193 true degree in its library, predictions are endpoint states, and pendulum-like data may appear in
 194 foundation-model corpora.

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A Prompt templates

The prompts used for every LLM evaluation cell are reproduced below verbatim.

Table 3: Inputs available to each family. State is at $t=0$; history is pre- $t=0$. Chronos is history-conditioned. *In changed_hidden cells the learned models receive normal-regime prior constants instead of the true ones.

Family	State	History	Constants	Equations
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ base; withheld*	— (learned)
Time-series (Chronos)	—	✓ (10 s context)	—	—
LLMs (kimi/grok/deepseek)	✓	—	except hidden	—

241 System prompt (direct / no-CoT).

242 You are a careful physicist predicting the future state of a k -pendulum (a chain of
 243 k point masses on massless rods, hinged in a planar chain, swinging under gravity
 244 with viscous damping). Respond with ONLY a single JSON object on one line, no prose,
 245 no code fences. Schema:
 246 {"theta": [...k floats...], "omega": [...k floats...]}
 247 theta_i is in radians (link i measured from straight-down,
 248 positive counter-clockwise)
 249 and omega_i is angular velocity in rad/s.

250 **System prompt (chain-of-thought).** Same preamble, then: “Reason step-by-step about the dynamics,
 251 then output your final answer. Wrap the final answer in `<answer>...</answer>` tags containing ONLY a
 252 single JSON object: {"theta": [...k floats...], "omega": [...k floats...]}.”

253 User prompt (coordinate modality).

254 Number of links $k = \{k\}$
 255 Physical constants:
 256 g (gravity, m/s^2): $\{g\}$
 257 L (rod lengths, m): $\{L\}$
 258 m (point masses, kg): $\{m\}$
 259 damping (viscous coefficient on omega): $\{\text{damping}\}$
 260 Initial state at $t=0$:
 261 θ_0 (rad): $\{\theta_0\}$
 262 ω_0 (rad/s): $\{\omega_0\}$
 263 Predict the state at $t = \{T\}$ seconds.
 264 Return theta and omega as length- $\{k\}$ JSON arrays.

265 **Hidden-constants regime.** The constants block above is replaced by:

266 Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point
 267 masses, and damping are NOT disclosed. You must infer dynamics from the initial
 268 conditions alone.

269 **Parsing and scoring.** Each response is stripped of code fences and (for CoT) of everything outside the
 270 `<answer>...</answer>` tags; the first `{...}` object is JSON-parsed and validated to contain length- k theta
 271 and omega arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

272 B Training and compute

273 All primary learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate, trained
 274 on a single 8 GB consumer GPU. Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN
 275 mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos
 276 runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at
 277 $\Delta t = 0.01$ s from the 10^{-3} s ground truth. Despite the number of samples, the training set contains only *nine*
 278 trajectories, one for each (k, regime) cell. The budget study uses nested subsets of 1, 5, and 20 trajectories per
 279 eligible cell and three training initializations. Neural ODE therefore sees 9, 45, or 180 trajectories; HNN sees 3,
 280 15, or 60 undamped trajectories. Evaluation uses the same two held-out seeds throughout.

281 **Training objectives.** Derivative matching minimizes MSE on finite-difference $\dot{s}$; rollout loss minimizes
 282 MSE on the state after k Euler steps with gradients through the unroll; mixed- k rollout trains on several window

lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$). The LNN’s loss requires second-order autodifferentiation and is trained at full scale (67k samples, hidden 256) on a CUDA GPU. Chronos-2 resamples the context to up to 512 points and forecasts ≤ 64 steps.

Reproducibility records. The anonymized artifact accompanying this submission contains the code, prompts, trained weights, held-out specifications, and the per-cell records for the main held-out comparison (2,160 records), the matched disclosure test (1,800), the modality and prompting grid, and the Lorenz sweep; the two large sweeps, the training-budget and balanced-seed studies, are packaged as separate archives. All dataset and held-out seeds are fixed and recorded in the released configuration. LLMs are sampled once at temperature 0. The reported hosted-model numbers recompute from the saved records; re-calling the API is a new evaluation rather than a reproduction. Invalid cells score $\pi/2$.

C Seed and system replication

Lorenz uses $(\sigma, \rho, \beta) = (10, 28, 8/3)$, RK4 ground truth at $\Delta t = 0.01$ s, a 5 s context, one fixed training seed, and five fixed held-out seeds (all recorded in the released configuration). Each held-out seed contains five trajectories, giving 25 held-out trajectories in total. The Neural ODE is a two-hidden-layer, width-128 tanh network trained for 300 epochs by analytic derivative matching. SINDy uses a degree-two polynomial library with STLSQ, fit to analytic derivatives at the sampled training states; the refit selects the true sparsity pattern exactly and recovers $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ with maximum absolute coefficient error 2.2×10^{-13} . NVAR uses two delays, degree-two features, and ridge regression. Chronos-2 uses `amazon/chronos-2` in univariate and multivariate modes.

Table 4: Replication results. Top: reliability-adjusted pendulum error (rad) by held-out seed and equal-seed mean; horizons $\{1, 10\}$ s. The original seed has 20 trajectories per (k , regime) cell and the new seed has five. Equal-seed answer rates are 97% for LNN, 61% for SINDy, and 46% for NVAR; all other rows answer 100%. Bottom: Lorenz–63 normalized MAE, averaged within each of five held-out seeds (five trajectories each) and then across seeds, with the standard error over seeds. SINDy is below 2×10^{-5} in every cell. Lower is better.

Pendulum model	primary seed	second seed	seed mean
rk4 (reference)	0.059	0.074	0.067
symplectic	0.205	0.194	0.200
euler	0.405	0.415	0.410
chronos2-uni	0.697	0.604	0.650
chronos2-multi	0.684	0.621	0.653
sindy	0.860	0.906	0.883
hnn-rollout	0.972	0.858	0.915
neural-ode-rollout-mixed	1.036	0.989	1.012
hnn-rollout-mixed	1.066	1.037	1.051
nvar	1.245	1.251	1.248
lnn	1.356	1.376	1.366
Lorenz model	0.1 s	0.5 s	1.0 s
rk4 (reference)	0.000	0.000	0.000
sindy	$<10^{-5}$	$<10^{-5}$	$<2 \times 10^{-5}$
neural-ode	0.014 (0.008)	0.079 (0.030)	0.277 (0.063)
nvar	0.012 (0.001)	0.262 (0.050)	0.429 (0.084)
chronos2-uni	0.126 (0.027)	0.700 (0.102)	0.732 (0.085)
chronos2-multi	0.101 (0.018)	0.691 (0.129)	0.771 (0.097)
euler	0.028 (0.005)	0.216 (0.023)	0.885 (0.120)
persistence	0.524 (0.054)	1.028 (0.123)	0.900 (0.057)

D Related work

The audited models span learned dynamics [Chen et al., 2018, Greydanus et al., 2019, Cranmer et al., 2020] on chaotic testbeds [Gilpin, 2021], zero-shot LLM and foundation-model forecasting [Gruver et al., 2023, Zhang and Gilpin, 2025, Ansari et al., 2024, Das et al., 2024, Woo et al., 2024], LLM physical reasoning [Lewkowycz et al., 2022, Wei et al., 2022, Xu et al., 2025], and sparse identification [Brunton et al., 2016, Gauthier et al., 2021]. The identifiability check draws on structural identifiability under parameter symmetries [Borgqvist et al.,

Table 5: Real-data check: mean wrapped angle error (rad) over both links on trials 2–3 (120 held-out states), failures at $\pi/2$. The physics rows use the published constants; “damped” adds the Trial-1-fitted viscous coefficient. Tracking noise floor ≈ 0.006 rad.

Model	0.1 s	0.5 s	1.0 s	Answered
physics-rk4	0.084	0.820	1.248	100%
physics-rk4-damped	0.086	0.892	1.145	100%
nearest-neighbor (cross-trial)	0.371	0.849	1.117	100%
chronos2	0.611	1.160	1.223	89%
constant-velocity	0.345	1.402	1.576	100%
sindy-real	0.490	1.501	1.535	99%
persistence	0.934	1.654	1.230	100%

2024]. Unlike selective prediction, which reports risk together with coverage [Geifman and El-Yaniv, 2017], failed cells here stay in the main score because models are compared at fixed task coverage.

E Real-data check details

The hardware dataset [Myers et al., 2020] is released under GPL-3 with video tracking of both link angles at 500 Hz and per-frame uncertainty estimates (median 0.20° upper link, 0.43° lower). We use the published simplified point-mass parameters ($L = [0.172, 0.143]$ m, $m = [0.311, 0.111]$ kg, $g = 9.81$). Angular velocities come from a Savitzky–Golay filter (window 42 ms, cubic, analytic derivative). The recordings begin with the pendulum positioned by hand, so the free drop is detected as the first time either link exceeds 4 rad/s; evaluation states are drawn evenly from 0.2 to 24 s after the drop, 60 per trial. The tracker zeroes its angle at the first frame, leaving a gravity-referenced offset per link that we select from $\{0, \pi\}$ by which choice lets the physics model track at a 20 ms horizon; the selected offsets are zero for every trial, with 20 ms error 0.005–0.007 rad against a 0.006 rad tracking floor. The damping grid $\{0, 5 \times 10^{-4}, \dots, 2 \times 10^{-2}\}$ selects 10^{-3} on Trial 1 at the 0.5 s horizon. SINDy uses a degree-two polynomial plus first-order Fourier library on Trial 1 with Savitzky–Golay derivatives; Chronos-2 receives a 5 s context resampled to 50 Hz and forecasts at most 50 steps. Failures score $\pi/2$; Chronos-2’s are input-infeasible cells (insufficient context) rather than divergences, and the fixed-coverage rule charges both failure modes alike.

F Extended diagnostics

This appendix expands the diagnostics summarized in §4.6.

Behavior by horizon. Out-of-sample error by horizon appears in Table 13. At 1 s the ordering follows the main comparison: the integrators are near-exact, kimi (0.457) has the lowest black-box error, and the learned models are at 0.86–1.25. At 10 s all models are in the chaotic regime, and the ordering is stable: kimi (0.619) remains lowest among black-box methods while the learned models reach 1.09–1.46. The mixed- k Neural ODE is mid-pack at both horizons (0.951 / 1.120). Figure 1 shows the growth curves over the $\{0.01, 1, 10, 60\}$ s ladder: the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The $\{1, 10\}$ s window used for the comparison is the range over which the families remain separable.

Breakdown by k and regime. Tables 6 and 7 report out-of-sample error by number of links and by regime. Error increases with k for every model except the mixed- k Neural ODE, which dips at $k=3$ (for kimi, $0.11 \rightarrow 0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family ordering holds within each k . The `changed_hidden` regime has the lowest error for almost every model (for kimi, 0.46 hidden vs. 0.64 normal). For the learned models, withholding the hidden constants reduces but does not remove the same difficulty effect: hidden-regime errors are now 0.71–1.19; the previous values favored models given the true constants. The low-gravity hidden constants ($g=1.62$) produce slower, less-divergent motion over a fixed horizon, so the unadjusted lower hidden-regime error contrast is a difficulty confound (§4.2). A cluster bootstrap over complete held-out trajectories changes no confidence interval by more than ~ 0.03 rad and leaves the ranking unchanged.

Training-objective check. Table 11 reports the effect of the loss with architecture fixed. Under the equalized hidden-constant analysis, rollout loss is not uniformly better: mixed- k rollout is lower for the Neural ODE, while single- k rollout is lower for the HNN. Full-scale CUDA training of the LNN gives the highest learned-model error (1.356), with 6% of cells diverging or exceeding the rollout timeout after memory-safe recovery. We make

Table 6: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (reference)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
grok-4-1-fast-reasoning	0.366	1.030	1.247
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout	0.524	0.985	1.406
neural-ode-rollout-mixed	0.845	1.178	1.084
hnn-rollout-mixed	0.550	1.162	1.487
neural-ode	0.626	1.304	1.400
neural-ode-rollout	0.968	1.172	1.224
hnn	0.511	1.337	1.567
lnn	1.072	1.440	1.556

no architecture-wide claim about LNNs; under our implementation and protocol the LNN was numerically unstable and generalized poorly out of sample.

Physical-fidelity check. Hamiltonian and Lagrangian networks target structure preservation, so we evaluate energy drift on the undamped (normal) regime, where true energy is constant. Table 12 reports mean absolute drift over 10 s, normalized by $E^* = g \sum_i m_i L_i$, and the fraction of bounded rollouts. The learned models do not match classical integrators on this axis: the lowest-drift learned model is 0.78, compared with 0.066 for the symplectic integrator and 0.494 for Euler. Symplectic rollout of the HNN leaves drift nearly unchanged, so changing the integrator does not account for the HNN’s drift. At the derivative-matching objective, HNN drift is comparable to Neural ODE drift (4.27 vs 4.35); the LNN is unstable, with 27% bounded rollouts. These results use the same nine-trajectory learned-model budget as the main comparison.

Multivariate time-series check. On 720 paired samples, Chronos-2 multivariate and univariate errors are indistinguishable: 0.543 (95% CI [0.498, 0.589]) versus 0.549 ([0.505, 0.593]). No effect remains after Holm correction, so the larger study rejects the apparent per- k trend in the primary grid.

Predictability horizon. Pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold τ from $t=0$; the contiguous rule is used because, after divergence, a later horizon can fall below τ by angle-wrap coincidence. Table 14 sweeps τ .

G Breakdowns by link count and regime

Reliability-adjusted mean angle error (rad) on the shared out-of-sample held-out grid (horizons $\{1, 10\}$ s), by number of links (Table 6) and by regime (Table 7); models in leaderboard order. Cells a model never produced (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$. Each row uses the same grid as Table 2.

H Additional result tables

Full per-model values for the horizon breakdown and the predictability horizon (both defined in Appendix F), together with the regime definitions, the matched disclosure gap, and the training-budget, training-objective, and physical-fidelity studies. The corresponding summary statistics appear in the main text.

Table 7: Out-of-sample error by regime. Learned-model hidden cells use the equalized no-hidden-constants run. The hidden regime remains easier for several models because its low-gravity constants give slower, less-divergent motion. The unadjusted regime contrast in §4.2 consequently mixes disclosure with difficulty.

Model	normal	disclosed	hidden
rk4 (reference)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
grok-4-1-fast-reasoning	1.096	0.959	0.588
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout	1.087	1.115	0.713
neural-ode-rollout-mixed	1.260	0.836	1.011
hnn-rollout-mixed	1.244	1.172	0.782
neural-ode	1.221	1.095	1.014
neural-ode-rollout	1.281	0.894	1.189
hnn	1.262	1.303	0.850
lnn	1.525	1.416	1.127

Table 8: Matched disclosure gap (rad), paired on identical trajectories. $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected. This experiment uses its own matched grid (3 models $\times$ 2 horizons $\times$ 150 trajectories = 900 pairs), not the shared held-out set, so its absolute errors are not comparable to Table 7. Per-model Δ values are computed before rounding.

Model	disclosed	hidden	Δ_{disc}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

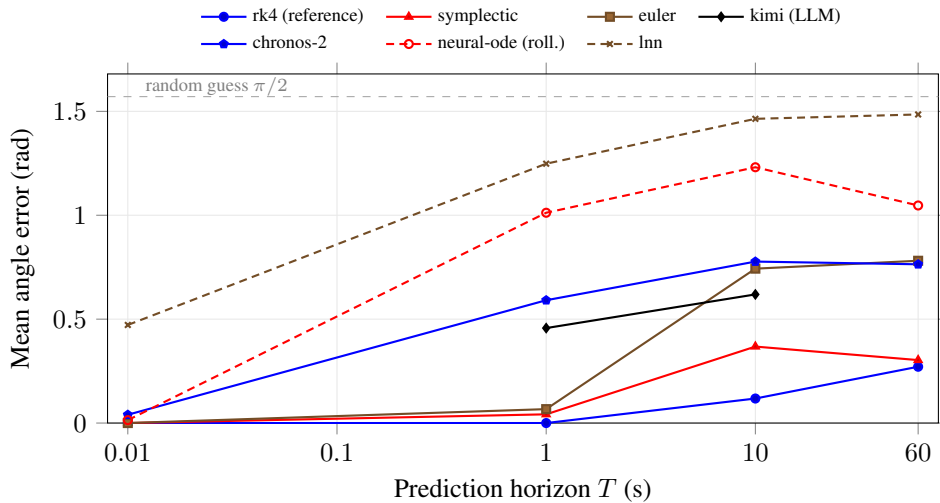


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs were evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 9: Physical regimes. `changed_hidden` withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
<code>changed_disclosed</code>	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
<code>changed_hidden</code>	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

Table 10: Nested training-budget study. Reliability-adjusted pendulum error (rad), averaged over three training initializations and two held-out seeds. Intervals hierarchically resample initializations and held-out seeds. HNN is evaluated on the undamped normal regime only.

Model	Scope	Trajectories	Error	95% CI
HNN	normal	3	1.269	[1.209, 1.349]
HNN	normal	15	0.974	[0.891, 1.062]
HNN	normal	60	0.981	[0.856, 1.113]
Neural ODE	all regimes	9	1.090	[1.011, 1.207]
Neural ODE	all regimes	45	0.968	[0.923, 1.028]
Neural ODE	all regimes	180	0.861	[0.829, 0.907]

Table 11: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.110	—
	rollout (single- k)	1.121	n.s. ($p_{\text{Holm}}=0.188$)
	mixed- k	1.036	sig. ($p_{\text{Holm}}=0.013$)
HNN	derivative MSE	1.138	—
	rollout (single- k)	0.972	sig. ($p_{\text{Holm}}=1.95 \times 10^{-4}$)
	mixed- k	1.066	sig. higher ($p_{\text{Holm}}=0.014$)

Table 12: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^*=g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. HNN drift is nearly unchanged when its rollout uses a symplectic integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

Table 13: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (reference)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimik2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
grok-4-1-fast-reasoning	0.905	0.857
deepseek-v4-pro	0.996	0.909
hnn-rollout	0.855	1.088
neural-ode-rollout-mixed	0.951	1.120
hnn-rollout-mixed	0.982	1.150
neural-ode	0.892	1.328
neural-ode-rollout	1.012	1.231
hnn	1.052	1.224
lnn (full-scale)	1.248	1.464

Table 14: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better; “—” means no horizon, including the first, stayed below τ .

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (reference)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	0.01
neural-ode-rollout-mixed	0.01	0.01	0.01	1.0
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	0.01
lnn	—	—	0.01	0.01